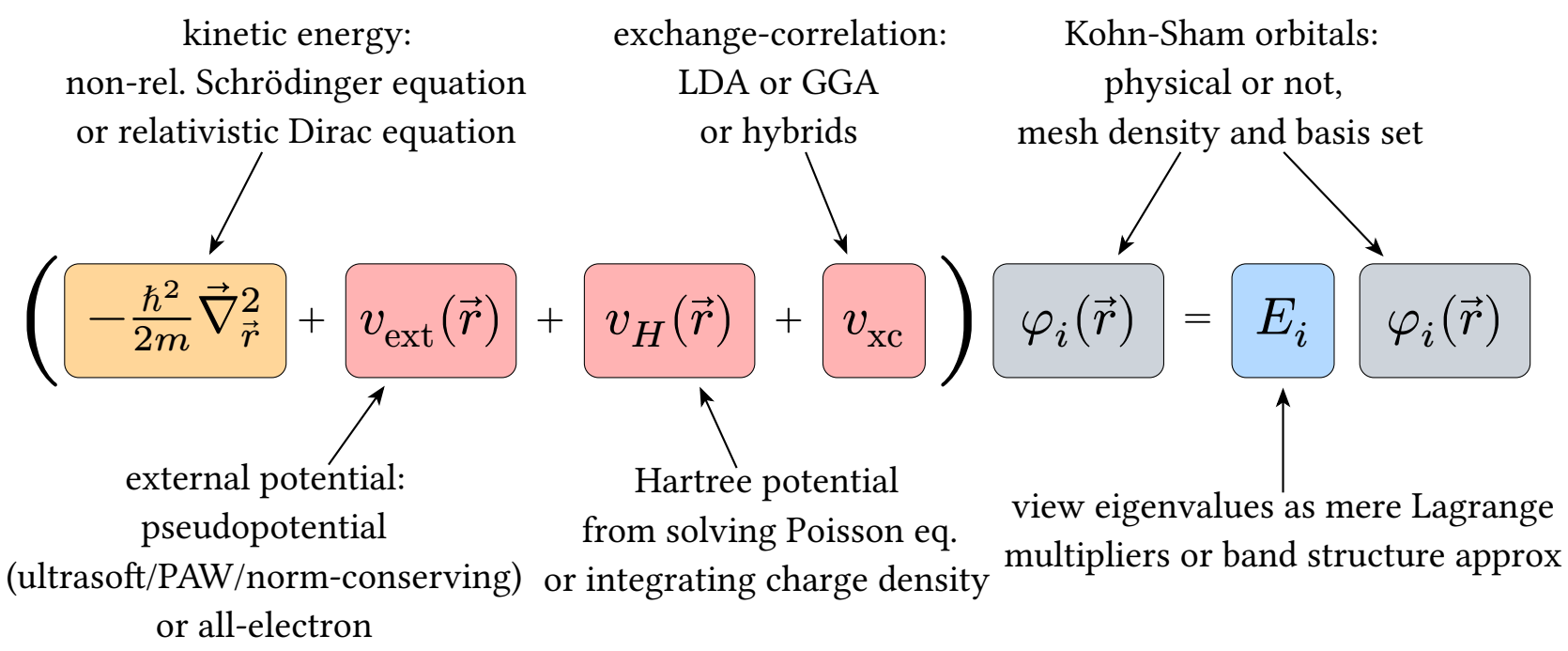
kinetic energy:
non-rel. Schrödinger equation
or relativistic Dirac equation
exchange-correlation:
LDA or GGA
or hybrids
Kohn-Sham orbitals:
physical or not,
mesh density and basis set
$\left(-\frac{\hbar^2}{2m}\vec{\nabla}^2_{\vec{r}} + v_{\text{ext}}(\vec{r}) + v_H(\vec{r}) + v_{\text{xc}}\right)\varphi_i(\vec{r}) = E_i\,\varphi_i(\vec{r})$
external potential:
pseudopotential
(ultrasoft/PAW/norm-conserving)
or all-electron
Hartree potential
from solving Poisson eq.
or integrating charge density
view eigenvalues as mere Lagrange
multipliers or band structure approx